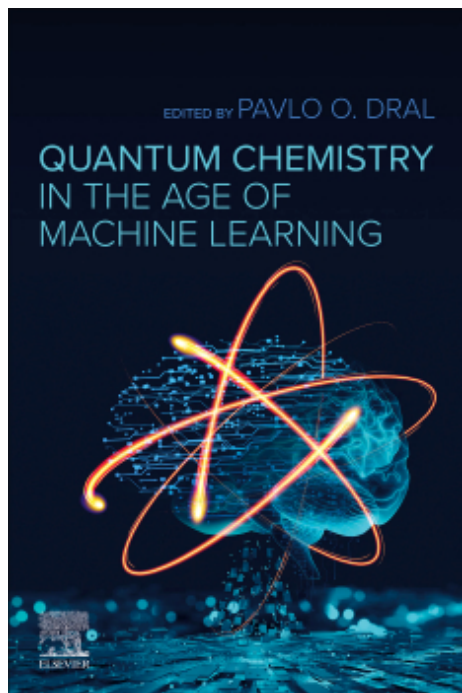




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Edition 1

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25. Machine learning wavefunction ↗
26. Analysis of nonadiabatic molecular dynamics trajectories ↗

27. Design of Organic Materials with Tailored Optical Properties: Predicting Quantum-Chemical Polarizabilities and Derived Quantities ↗

Material for Case studies

Part 1. Introduction

Chapter 1. Very brief introduction to quantum chemistry by Xun Wu and Peifeng Su

<https://github.com/dralgroup/MLinQCbook22-CH01> ↗

Chapter 2. Density functional theory by Hong Jiang and Huai-Yang Sun

<https://github.com/ffshy/ChapterDFTCaseStudy> ↗

Chapter 3. Semiempirical quantum mechanical methods by Pavlo O. Dral and Jan Řezáč

<https://github.com/dralgroup/MLinQCbook22-SQM> ↗

Chapter 4. From small molecules to solid-state materials: A brief discourse on an example of carbon compounds by Bili Chen, Leyuan Cui, Shuai Wang, Gang Fu

<https://github.com/bilio501/MLinQCbook22-CH04> ↗

Chapter 5. Basics of dynamics by Xinxin Zhong and Yi Zhao

<https://github.com/Cindy611/TDQD> ↗

Chapter 6. Machine learning: An overview by Eugen Hruska and Fang Liu

<https://github.com/Liu-group/MLbook> ↗

Chapter 7. Unsupervised learning by Rose K. Cersonsky and Sandip De

<https://github.com/rosecers/unsupervised-ml> ↗

Chapter 8. Neural networks by Pavlo O. Dral, Alexei Kananenka, Fuchun Ge, Bao-Xin Xue

<https://github.com/dralgroup/MLinQCbook22-NN> ↗

Chapter 9. Kernel methods by Max Pinheiro Jr and Pavlo O. Dral

<https://github.com/dralgroup/MLinQCbook22-NN> ↗

Chapter 10. Bayesian inference by Wei Liang and Hongsheng Dai

<https://github.com/WeiLiangXMU/Bayesian-Inference> ↗

Part 2. Machine learning potentials

Chapter 11. Potentials based on linear models by Gauthier Tallec, Gaétan Laurens, Owen Fresse-Colson, Julien Lam

<https://github.com/julienlamcnrs/Exercices-Potentials-based-on-linear-models.git> ↗

Chapter 12. Neural network potentials by Jinzhe Zeng, Liqun Cao, Tong Zhu

<https://github.com/tongzhugroup/Chapter13-tutorial> ↗

Chapter 13. Kernel method potentials by Yi-Fan Hou and Pavlo O. Dral

<https://github.com/dralgroup/MLinQCbook22-KMP> ↗

Chapter 14. Constructing machine learning potentials with active learning by Cheng Shang and Zhi-Pan Liu

www.lasphub.com/supportings/Li-GMsearch-AL.tgz ↗

Chapter 15. Excited-state dynamics with machine learning by Lina Zhang, Arif Ullah, Max Pinheiro Jr, Mario Barbatti, Pavlo O. Dral

<https://github.com/maxjr82/MLinQCbook16-NAMD> ↗

Chapter 16. Machine learning for vibrational spectroscopy by Sergei Manzhos, Manabu Ihara, Tucker Carrington

<https://github.com/sergeimanzhos/QCAML> ↗

Chapter 17. Molecular structure optimizations with Gaussian process regression by Roland Lindh and Ignacio Fernández Galván

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Part 3. Machine learning of quantum chemical properties

Chapter 18. Learning electron densities by Bruno Cuevas-Zuviría

<https://github.com/brunocuevas/density-learning-tutorials> ↗

Chapter 19. Learning dipole moments and polarizabilities by Yaolong Zhang, Jun Jiang, Bin Jiang

<https://github.com/zylustc/Learning-Dipole-Moments-and-Polarizabilities> ↗

Chapter 20. Learning excited-state properties by Julia Westermayr, Pavlo O. Dral, Philipp Marquetand

Case study 1

<http://mlatom.com/mlinqcbook22-mlesprops/> ↗

Case study 2

Code and tutorial: https://github.com/schnarc/SchNarc/tree/DipoleMoments_Spectra ↗

Data: <https://bit.ly/3lnUaZb> ↗

Part 4. Machine learning-improved quantum chemical methods

Chapter 21. Learning from multiple quantum chemical methods: Δ -learning, transfer learning, co-kriging, and beyond by Pavlo O. Dral, Tetiana Zubatiuk, Bao-Xin Xue

<https://github.com/dralgroup/MLinQCbook22-delta> ↗

Chapter 22. Data-driven acceleration of coupled-cluster and perturbation theory methods by Grier M. Jones, P. D. Varuna S. Pathirage, Konstantinos D. Vogiatzis

Code examples of the case studies: https://ChemRacer.github.io/DDQC_Demo/ ↗ Source code: https://github.com/ChemRacer/DDQC_Demo ↗

Chapter 23. Redesigning density functional theory with machine learning by Jiang Wu, Guanhua Chen, Jingchun Wang, Xiao Zheng

<https://github.com/zhouyyc6782/oep-wy-xcnn> ↗

Chapter 24. Improving semiempirical quantum mechanical methods with machine learning by Pavlo O. Dral and Tetiana Zubatiuk

Initial guess for the ethylene geometry:

6

C	-0.723601672	0.000000000	-1.235611088
C	-0.723601672	0.000000000	0.094546912
H	-0.723601672	0.923341000	-1.808561088
H	-0.723601672	-0.923341000	-1.808561088
H	-0.723601672	0.923341000	0.667496912
H	-0.723601672	-0.923341000	0.667496912

Follow the instructions at <http://mlatom.com/AIQM1> ↗ to perform geometry optimization and thermochemical calculations with AIQM1.

Chapter 25. Machine learning wavefunction by Stefano Battaglia

<https://github.com/stefabat/MLWavefunction> ↗

Part 5. Analysis of Big Data**Chapter 26. Analysis of nonadiabatic molecular dynamics trajectories by Yifei Zhu, Jiawei Peng, Hong Liu and Zhenggang Lan****Case study 1**

https://figshare.com/articles/dataset/Case_study_1_Classical_MDS_analysis_of_CH2NH2_dynamics/17110610 ↗

Case study 2

https://figshare.com/articles/dataset/Case_study_2_Fr_chet_distance_analysis_of_phytochromobilin/17104457 ↗

Case study 3

https://figshare.com/articles/dataset/Case_study_3_PCA_of_site-exciton_model_dynamics/17110592 ↗

Chapter 27. Design of organic materials with tailored optical properties: Predicting quantum-chemical polarizabilities and derived quantities by Gaurav Vishwakarma, Aditya Sonpal, Aatish Pradhan, Mojtaba Haghighatlari, Mohammad Atif Faiz Afzal, Johannes Hachmann

Code snippets are provided directly in the chapter text.

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